

Energy-Efficient UAV–IRS Coordination in Vehicular Networks Using GNN-Enhanced MADRL

Laleh Eslami, Behrouz Shahgholi Ghahfarokhi, Mohammad Reza Khayyambashi, and Pin-Han Ho



Abstract—This paper investigates the joint optimization of unmanned aerial vehicle (UAV) deployment, intelligent reflecting surface (IRS) configuration, and user association in vehicular networks, aiming to maximize downlink energy efficiency under dynamic large-scale topologies. The problem is formulated as a mixed-integer non-convex program that captures coupled UAV positioning, zero-forcing (ZF) beamforming, IRS phase-shift tuning, and user scheduling, subject to mobility and power constraints. To tackle its complexity, we propose a hybrid iterative framework: UAV beamforming is optimized via ZF precoding, while the remaining variables are updated using a graph neural network (GNN)-enhanced multi-agent deep reinforcement learning (MADRL) scheme under centralized training with decentralized execution (CTDE). The GNN captures spatial-relational dependencies among UAVs, IRS panels, and vehicles, enabling scalable policy learning in high-mobility environments. Simulation results on realistic vehicular mobility traces reveal that the proposed method achieves notable gains in energy efficiency and convergence speed compared to conventional MARL and optimization-based baselines.

Index Terms— vehicular network, intelligent reflecting surface (IRS), unmanned aerial vehicles (UAV), MADRL, CTDE.

I. INTRODUCTION

The integration of unmanned aerial vehicles (UAVs) and intelligent reflecting surfaces (IRS) into vehicular networks has opened new frontiers for intelligent transportation systems (ITS). UAVs can be rapidly deployed to extend coverage, deliver high-capacity links, and adapt to dynamic user distributions in complex urban topologies. When coupled with IRS, the system can manipulate the wireless environment via passive phase reconfiguration, thereby enhancing spectral and energy efficiency without requiring additional active power sources [1].

Reinforcement learning (RL) enables an agent to learn optimal sequential decision-making policies through interaction with an environment, aiming to maximize long-term cumulative rewards. Owing to its ability to operate without explicit system models, RL is well suited for dynamic and partially unknown environments [2].

In wireless networks, tightly coupled variables such as power control, beamforming, and node positioning evolve under stochastic conditions, making conventional optimization methods with static assumptions impractical [3]. RL provides adaptive control through online interaction, addressing these challenges effectively.

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Deep reinforcement learning (DRL) extends RL to large or continuous state and action spaces by employing deep neural networks. Representative methods include DQN for discrete actions and DDGP for continuous control using actor–critic architectures [4].

Multi-agent reinforcement learning (MARL) generalizes RL to multi-entity systems, where agents' actions and rewards are interdependent [5], which is essential for UAV-IRS-assisted vehicular networks. A common MARL framework is centralized training with decentralized execution (CTDE), which stabilizes learning using global information during training while enabling fully distributed operation during execution [6].

However, the control and optimization of UAV-IRS-assisted vehicular networks is inherently challenging due to the coupled nature of spatial placement, beamforming, IRS phase tuning, and user association. The problem is typically formulated as a mixed-integer non-convex optimization, whose global optimum is computationally prohibitive to derive in real-time large-scale networks. This complexity is compounded by high mobility and frequently changing channel conditions in Internet-of-Vehicles (IoV) scenarios.

For UAV-IRS cooperation, CTDE is especially advantageous: UAVs can learn coordinated movement and beamforming strategies, while IRS controllers learn optimal reflection configurations, all under a shared objective function. The centralized training phase can capture the full system topology and channel state, while decentralized execution ensures that, in real asynchronous and resource-constrained deployments, each agent can react promptly to local conditions without waiting for global updates.

In the emerging field of UAV-IRS-assisted communications, researchers have investigated joint aerial platform positioning and reflective surface configuration to cope with rapid topology changes and non-line-of-sight challenges. The authors in [7] propose a joint optimization framework combining IRS-assisted passive beamforming and power allocation to maximize system capacity under NOMA. While effective for improving spectral efficiency in urban scenarios, it ignores UAV trajectory and active beamforming, which are crucial in dynamic multi-user settings. In [8], the authors study a NOMA-based multi-UAV network assisted by multiple IRSs, where ground user association, UAV transmit power, and IRS phase shifts are jointly optimized to maximize spectral efficiency under imperfect successive interference cancellation. The proposed framework adopts a centralized, model-based optimization approach and focuses on static or quasi-static network scenarios, without considering UAV mobility dynamics or energy-efficiency-oriented objectives. In the model introduced in [9], the IRS phase shift coefficients and the UAV trajectory are optimized to maximize the average bit rate of vehicles in both UAV-to-vehicle and vehicle-to-vehicle communications.

MARL and DRL have been extensively applied in UAV and IRS-assisted wireless networks to address high-dimensional, decentralized control under channel uncertainty, interference, and mobility. The authors in [10] investigated joint resource allocation and transmit beamforming in cellular-connected UAV networks. Their hybrid deep double dueling Q network (D3QN) and twin delayed deep deterministic policy gradient (TD3) framework shows the effectiveness of learning-based methods for coupled radio resource management and interference mitigation. In joint UAV trajectory and IRS phase-shift optimization, studies have employed double deep Q-network (DDQN), DDPG, long short-term memory (LSTM)-DDPG, and hybrid DRL–neural mapping models to improve metrics such as propulsion-energy efficiency [11], downlink rate [4], and secrecy capacity [12], [13]. These methods enable adaptive path planning and passive-beam reconfiguration without explicit channel models, achieving faster convergence and notable performance gains compared to alternating optimization and single-agent RL. In addition, the work in [14] introduces an integrated architecture that combines UAVs, unmanned ground vehicles (UGV)-mounted RIS units, and wireless power transfer to support continuous service delivery. The framework employs a MARL strategy to coordinate UAV and UGV mobility, enable energy-efficient operation, and perform probabilistic content caching based on request popularity.

For integrated sensing and communication (ISAC), multi-agent frameworks such as random network distillation-enhanced multi-agent deep deterministic policy gradient (RND-MADDPG) [15] and hierarchical-reward multi-agent proximal policy optimization (MAPPO) [5] jointly enhance sensing fidelity and throughput, while hierarchical model-based designs further optimize multi-UAV RIS-assisted ISAC under structured channels [16].

In UAV-IRS-NOMA and UAV-assisted vehicular edge computing (VEC), the optimization objectives extend to spectral-efficiency enhancement, task-offloading schedules, and decoding-order control. Distributed MARL variants, such as decaying DQN [17], multiagent twin delayed deep deterministic policy gradient (MATD3) [18], MADDPG with capability-aware coordination [20], mean-field game-aided actor-critic [20], and distributed many-agent learning frameworks for UAV-assisted MEC [21] reduce communication overhead, allow large-scale agent deployment, and improve energy efficiency and latency. Cross-domain integrations such as cell-free massive MIMO or device-to-device (D2D) communications with UAV-IRS [22], [23] employ DRL-driven scheduling to improve RF energy harvesting, prolong IoT device lifetime, and preserve QoS in heterogeneous networks.

Graph neural networks (GNNs) have been utilized for beamforming, UAV placement, and freshness-oriented trajectory optimization, where message passing captures spatial correlation and interference among nodes. Two-stage residual GNNs for hybrid maximum ratio transmission (MRT)/zero-forcing (ZF) beamforming [24], CTDE-enabled GNN trajectory planning for age of information (AoI) minimization [25], and hierarchical GNN/soft actor-critic designs for secure beamforming [26] demonstrate scalability and permutation equivariance across multi-UAV deployments. A separate line of research explicitly leverages the CTDE paradigm, wherein agents are jointly trained with global

network knowledge and later operate using local observations. Works in UAV trajectory planning [25], ISAC optimization [16], cooperative multi-UAV scheduling [27], and distributed state-action-reward-based user association [28] use CTDE to accelerate convergence, improve coordination, and maintain adaptability in partially observable, large-scale environments. Similarly, CTDE-aided distributed IRS precoding and user scheduling [2] achieves near-optimal sum-rate with drastic computation-time reduction. To better contextualize the proposed framework, Table I provides a comparative overview of state-of-the-art optimization-based and multi-agent learning approaches, highlighting their network models, state representations, scalability, and ability to adapt to dynamic environments, which motivates the design choices adopted in this work.

While prior works have advanced UAV placement, IRS control, and MARL-based resource allocation independently, few have addressed the joint, large-scale, and dynamic optimization of all three components in realistic vehicular environments. Notably, existing solutions often fail to capture the relational structure among network entities in each time slot, to scale effectively with large numbers of vehicles, UAVs, and IRS panels, and to sustain high energy efficiency under rapid topology changes. In related large-scale distributed systems, scalable coordination mechanisms have been explored through adaptive clustering [29]. However, such clustering-oriented approaches have not been extended to joint UAV-IRS-vehicular optimization scenarios. Moreover, in multi-agent settings, few works harness deep architectures capable of extracting rich representations from high-dimensional, graph-structured states, which limits their adaptability in complex scenarios. To overcome these challenges, this paper makes the following contributions:

- We formulate an optimization problem that jointly considers multi-UAV deployment, IRS configuration, user association, and beamforming in a large-scale vehicular network, capturing mobility constraints, power limits, and dynamic channel conditions in a mixed-integer non-convex form.
- To solve the formulated problem, we design a deep MARL-based architecture enhanced with a model-based ZF beamforming module, enabling sample-efficient, fast-converging, and scalable joint optimization of UAV positioning, IRS configuration, and user association in large vehicular networks, a combination not explored in prior UAV-IRS MARL literature.
- The UAV-IRS-vehicle network is represented as a dynamic graph, enabling a GNN to extract spatial-relational features for MARL policy learning.
- A custom reward function is designed to guide the CTDE process to balance spectral performance and energy consumption, ensuring long-term UAV endurance.

The remainder of this paper is structured as follows: Section II outlines the system model and problem formulation. In Section III, we introduce the proposed solution. Section IV presents numerical results and performance evaluation of the proposed model. Next, Section V discusses the computational complexity and signaling overhead of our framework, and finally, Section VI concludes the paper.

Table I. Comparative overview of state-of-the-art optimization-based and MARL approaches

Feature Ref.	Network Model	State Representation	Optimized Parameters	Optimization Approach	Scalability	Handling Dynamics
[2]	IRS-only	Local observations	User scheduling, BS precoding, IRS phase shifts	MADRL/CTDE	High	High
[14]	IRS-UAV	Global system information	Transmit power and association of users, UAV beamforming, IRS phase shifts	Hierarchical model-based	Medium	Medium
[21]	UAV-only	Graph-structured partial observations	Agent actions (policy-level resource control)	MARL/CTDE + GNN	High	High
[22]	UAV-only	Graph-structured state representation	Transmit power, link-level control variables	DRL/CTDE + GNN	High	Medium–High
[23]	UAV-only	History + neighbor info (embedding)	Policy parameters, meta-learning parameters, uncertainty	Meta offline MARL	Medium	Low
[24]	UAV-only	Local observations	UAV trajectory, user association	Cooperative MARL/CTDE	Medium	Medium–High
Proposed Work	IRS-UAV	Graph-structured partial observations	UAV positioning and beamforming, IRS phase-shifts, user association	MADRL/CTDE + GNN	High	High

Throughout this paper, bold lowercase letters represent vectors, while bold uppercase letters denote matrices. Notation $(\cdot)^T$ is used to represent the transpose of a vector or matrix. Besides, $\text{diag}(\cdot)$ denotes a diagonal matrix with the elements of a vector on the main diagonal, and $|\cdot|$ and $\|\cdot\|$ represent the modulus and Euclidean norm, respectively. Notation $\mathbb{C}^{a \times b}$ is the set of $a \times b$ complex matrix. Notation $\mathcal{CN}(\mu, \sigma^2)$ refers to the complex Gaussian distribution with mean μ and variance σ^2 . The 3-D distance between two network members a and b is defined as $d_{ab} = \sqrt{\|\Phi_a - \Phi_b\|^2}$, where $\Phi_a = [x_a, y_a, h_a]^T$ denotes the 3-D Cartesian coordinates of a , and $\phi_a = [x_a, y_a]^T$ represents its horizontal location.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this study, we consider the downlink communication in the network illustrated in Fig. 1. A set of U fixed-wing UAVs are deployed as aerial base stations (ABSs) for the network. Each UAV $u \in \mathcal{U} = \{1, \dots, U\}$ is equipped with M transmit/receive antennas and serves a set of single-antenna ground vehicles denoted by $\mathcal{N} \in \{1, \dots, N\}$.

To assist the communication, we consider a hotspot-oriented deployment scenario in which one IRS is mounted on the wall of the tallest building located within the coverage area of each UAV. The set of IRSs is denoted by $\mathcal{R} \triangleq \{1, \dots, r, \dots, R\}$, where IRS r is associated with UAV u . This association is determined at the beginning of each UAV mission by selecting the nearest IRS. The inter-IRS spacing is sufficiently small, ensuring that the IRS assigned to each UAV stays within its coverage area. Moreover, given the short time-slot duration and the limited UAV mobility, the geometric relationship between a UAV and its assigned IRS changes slowly, which justifies maintaining a fixed association during the mission. Each IRS consists of K passive reflecting elements, and the complex reflection coefficient of the k^{th} element is denoted by $\theta_k = e^{j\phi_k}$, with $\phi_k \in [0, 2\pi), \forall k \in \mathcal{K} \triangleq \{1, \dots, K\}$.

The three-dimensional Cartesian coordinates of the UAV u , vehicle n , and IRS r are denoted by $\Phi_u^{\text{UAV}} = [x_u, y_u, h_u]^T$,

$\Phi_n^{\text{V}} = [x_n, y_n, 0]^T$, and $\Phi_r^{\text{IRS}} = [x_r, y_r, h_r]^T$, respectively. The UAV flight altitude is assumed to be fixed, i.e., $h_u = H > 0$.

The coverage region is modeled as a circular area projected onto the ground with radius R_u . Specifically, a vehicle n is considered to be within the coverage of UAV u at time slot t if $\|\Phi_u^{\text{UAV}}[t] - \Phi_n^{\text{V}}[t]\| \leq R_u$. Each vehicle is assumed to be exclusively served by one UAV. To represent this association, we define the binary variable $\alpha_{u,n}$, where $\alpha_{u,n} = 1$ if vehicle n is associated with UAV u , and $\alpha_{u,n} = 0$ otherwise. All UAVs share the same frequency band to serve their associated vehicles. Therefore, the beamforming design at each UAV must be capable of mitigating the inter-UAV interference.

The system optimization is performed over a time horizon consisting of T discrete time slots, each with a duration of Δt . The slot duration Δt is chosen to be sufficiently small so that the channel gains can be assumed constant within each slot.

A. Channel Model

According to the 3GPP standard [30], the direct air-to-ground link between UAV u and vehicle n at time slot t is modeled as

$$\mathbf{h}_{un}[t] = \tilde{\mathbf{h}}_{un}[t] 10^{-L_{un}[t]/10}, \quad (1)$$

where $\mathbf{h}_{un}[t] \in \mathbb{C}^{1 \times M}$, $\tilde{\mathbf{h}}_{un} \sim \mathcal{CN}(0, \mathbf{I})$ represents Rayleigh fading and L_{un} is the path loss (in dB) is the path loss (in dB) determined according to the 3GPP model in [31]. Similarly, the channel between the IRS and vehicle n is modeled as

$$\mathbf{h}_{rn}[t] = \tilde{\mathbf{h}}_{rn}[t] 10^{-L_{rn}[t]/10}, \quad (2)$$

where $\mathbf{h}_{rn}[t] \in \mathbb{C}^{K \times 1}$, and $\tilde{\mathbf{h}}_{rn}[t] \sim \mathcal{CN}(0, 1)$ represents small-scale fading, and $L_{rn}[t]$ denotes the path loss determined according to the 3GPP model in [30].

We define the array response vector as

$$\mathbf{a}(\bar{\delta}, \bar{K}) = \frac{1}{\sqrt{\bar{K}}} [1, \dots, e^{-j\pi(\bar{K}-1)\bar{\delta}}], \quad (3)$$

where $\bar{\delta}$ denotes the cosine of the phase difference between the signals received by adjacent antenna elements in a uniform linear array (ULA) of $\bar{K}$ antennas, with the inter-element spacing equal to half the wavelength.

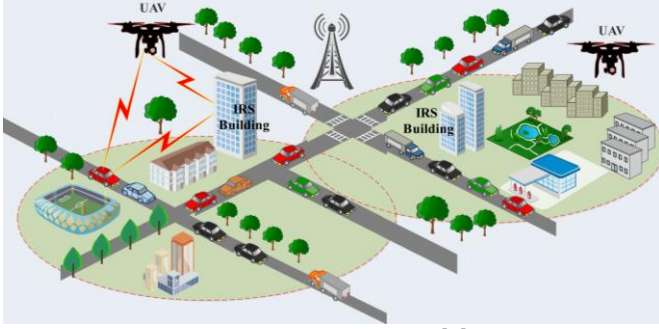


Fig. 1. System Model

Since the UAVs operate at high altitudes and the IRS is mounted on the façade of a building, the UAV–IRS link is assumed to be LoS. Accordingly, the channel matrix between UAV u and the IRS, $\mathbf{H}_{ur} \in \mathbb{C}^{K \times M}$, is expressed as

$$\mathbf{H}_{ur}[t] = \sqrt{MK\rho d_{ur}^{-\zeta}} \mathbf{a}^T(\bar{\delta}_{ur}, K) \otimes \mathbf{a}(\bar{\delta}_{ur}M), \quad (4)$$

where ρ denotes the path loss at the reference distance $d=1\text{m}$, ζ is the path-loss exponent, and $\bar{\delta}_{ur} = \frac{x_u - x_r}{d_{ur}}$ is the cosine of the angle of arrival (AoA) for the UAV–IRS signal.

B. Signal Model

Let $S_{u,n}[t]$ denote the downlink signal intended for vehicle n and generated by UAV u in time slot t with a duration of Δt . Accordingly, the downlink signal transmitted from UAV u to vehicle n is given by

$$\tilde{S}_{u,n}[t] = \mathbf{w}_{u,n}[t] S_{u,n}[t] \quad (5)$$

where $\mathbf{w}_{u,n}[t] \in \mathbb{C}^{M \times 1}$ is the corresponding transmit beamforming vector at time slot t . The downlink transmission is conveyed through both the direct channel and the IRS-assisted reflective channel. Therefore, the received signal at vehicle n is expressed as

$$y_{u,n} = \sum_{n' \in \mathcal{N}} \mathbf{h}_{u,n}^c[t] \mathbf{w}_{u,n'}[t] S_{u,n'}[t] + z_{u,n}[t] \quad (6)$$

where $z_{u,n}[t] \sim \mathcal{CN}(0, \sigma_z^2)$ denotes the additive white Gaussian noise (AWGN) at the receiver of vehicle n , and $\mathbf{h}_{u,n}^c[t]$ represents the composite downlink channel, defined as:

$$\mathbf{h}_{u,n}^c[t] = \mathbf{h}_{un}[t] + \mathbf{h}_{rn}^T[t] \mathbf{\Theta}_u[t] \mathbf{H}_{ur}[t], \quad (7)$$

where $\mathbf{\Theta}_u[t] = \text{diag}(\theta_{u,1}, \dots, \theta_{u,k}, \dots, \theta_{u,K}) \in \mathbb{C}^{K \times K}$ is the IRS reflection matrix, in which $\theta_{u,k}$ denotes the complex reflection coefficient of the k^{th} IRS element associated with UAV u . Accordingly, the achievable data rate at vehicle n in time slot t , normalized per unit bandwidth (Hz), is given by:

$$R_{u,n}[t] = \log_2 \left(1 + \frac{|\mathbf{h}_{u,n}^c[t] \mathbf{w}_{u,n}[t]|^2}{\sum_{n' \neq n} |\mathbf{h}_{u,n'}^c[t] \mathbf{w}_{u,n'}[t]|^2 + \sigma_z^2} \right), \quad (8)$$

C. Energy Consumption Model

The total energy consumption of a fixed-wing UAV comprises two main components: (i) communication-related energy consumption, and (ii) propulsion-related energy consumption. The first component accounts for transmission, signal processing, and circuit operations, whereas the second corresponds to the energy required to sustain aerodynamic lift and forward flight.

It is well established that, for fixed-wing UAVs, propulsion energy dominates the overall energy consumption and typically accounts for more than 95% of the total energy usage. Under this assumption, the total energy consumption of each UAV is well approximated by its propulsion energy consumption, i.e., $E_u^{\text{tot}} \approx E_u^{\text{fly}}$ [18]. For a fixed-wing UAV, the propulsion energy consumption is strongly dependent on the flight speed and can be modeled as [31]:

$$E_u^{\text{fly}} = \Delta t \left(\tau_1 V_u^3[t] + \frac{\tau_2}{V_u[t]} \right) \quad (9)$$

where $V_u[t] = \|\boldsymbol{\phi}_u^{\text{UAV}}[t] - \boldsymbol{\phi}_u^{\text{UAV}}[t-1]\|/\Delta t$ denotes the flight speed of UAV u in time slot t . The parameters τ_1 and τ_2 are determined by factors such as the UAV weight, wing area, wing aspect ratio, and air density [31].

D. Problem Formulation

The objective is to maximize the downlink energy efficiency over the UAV mission duration by jointly optimizing the user association parameters $\alpha_{u,n}, \forall u \in \mathcal{U}, \forall n \in \mathcal{N}$, the transmit beamforming vectors of each UAV, $\mathbf{W}_u, \forall u \in \mathcal{U}$, the reflection matrices of IRS panels, $\mathbf{\Theta}_u, \forall u \in \mathcal{U}$, and the horizontal positions of each UAV, $\boldsymbol{\phi}_u, \forall u \in \mathcal{U}$. Accordingly, the formulated optimization problem can be expressed as:

$$\begin{aligned} \mathcal{P1} \quad & \max_{\mathbf{A}, \mathbf{W}, \mathbf{\Theta}, \boldsymbol{\Phi}} \quad \eta_{ee} = \frac{\sum_{t=1}^T \sum_{u=1}^U \sum_{n=1}^N \alpha_{u,n}[t] R_{u,n}[t]}{\sum_{t=1}^T \sum_{u=1}^U E_u^{\text{tot}}[t]} \\ \text{s.t.} \quad & \text{C1: } \|\mathbf{W}_u\|_F^2 \leq P, \forall u \in \mathcal{U}, \\ & \text{C2: } \|\boldsymbol{\phi}_u[t] - \boldsymbol{\phi}_u[t-1]\|^2 \leq \delta_{\max}^2, \forall t \in \{1, \dots, T+1\}, \forall u \in \mathcal{U}, \\ & \text{C3: } \|\boldsymbol{\phi}_u^{\text{UAV}} - \boldsymbol{\phi}_{u'}^{\text{UAV}}\|^2 \geq \delta_{\min}^2, \forall u, u' \in \mathcal{U}, u \neq u', \\ & \text{C4: } |\theta_{u,k}| = 1, \forall k \in \{1, \dots, K\}, \forall u \in \mathcal{U} \\ & \text{C5: } \sum_{u=1}^U \alpha_{u,n} = 1, \forall n \in \mathcal{N}, \\ & \text{C6: } \sum_{n=1}^N \alpha_{u,n} \leq N_u, \forall u \in \mathcal{U}, \\ & \text{C7: } \alpha_{u,n} \in \{0, 1\}, \forall u \in \mathcal{U}, \forall n \in \mathcal{N}. \end{aligned}$$

Here, $\mathbf{A} \triangleq (\alpha_{u,n})_{u \in \mathcal{U}, n \in \mathcal{N}}$ denotes the UAV–vehicle association matrix, $\mathbf{W} \in \mathbb{C}^{U \times M \times N}$ represents the aggregated transmit beamforming matrix of all UAVs, and $\boldsymbol{\Phi}$ denotes the matrix of horizontal positions of all UAVs. The parameter P is the maximum transmit power allowed for each UAV. The quantity $\delta_{\max} = V_{\max} \Delta t$ represents the maximum flight distance of a UAV within a single time slot, where $V_{\max}$ is the maximum UAV speed. The parameter $\delta_{\min}$ denotes the minimum safety distance between the UAVs. The vectors $\boldsymbol{\phi}_u[t]$ and $\boldsymbol{\phi}_u[t-1]$ indicate the horizontal positions of UAV u in time slots t and $t-1$, respectively. In the formulated optimization problem, constraint C1 ensures that the transmit power of each UAV does not exceed P . Constraint C2 enforces the flight distance limit per UAV in each time slot. Constraint C3 prevents the collision between UAVs. Constraint C4 preserves full amplitude reflection from each IRS element and allowing only phase adjustments. This maximizes the reflected

power from every IRS element for constructive combining at the receiver. Constraint C5 ensures that each vehicle is associated with exactly one UAV, and constraint C6 limits the number of vehicles served by UAV u to N_u , where $N_u \leq M$ is required for ZF feasibility. Constraint C7 indicates that the association variable $\alpha_{u,n}$ should be binary-valued.

Problem P1 is a mixed-integer non-convex optimization problem with strongly coupled decision variables. The achievable rate $R_{u,n}[t]$ depends on the effective channel gain, which is jointly determined by the UAV horizontal position $\boldsymbol{\phi}_u$ and the IRS phase-shift matrix $\boldsymbol{\Theta}_u$. In particular, the UAV position affects the large-scale path loss and LoS probability, while $\boldsymbol{\Theta}_u$ modifies the cascaded UAV–IRS–vehicle channel and thus the resulting SINR. Meanwhile, the association variable $\alpha_{u,n}$ determine the set of users simultaneously served by each UAV, which directly influences the dimension and feasibility of the ZF beamforming matrix $\mathbf{W}_u$ (as enforced by constraint C6, where $N_u \leq M$). Furthermore, $\mathbf{W}_u$ is computed based on the instantaneous effective channel, which itself depends on $\boldsymbol{\phi}_u$ and $\boldsymbol{\Theta}_u$, creating an additional dependency between deployment, IRS configuration, and precoding design. Moreover, UAV mobility constraints restrict feasible position updates across time slots, while per-UAV transmit power limits constrain the beamforming design, further tightening the coupling among deployment, scheduling, and precoding variables in the mixed-integer non-convex formulation. Finally, the energy efficiency objective η_{ee} couples all variables through both the achievable rate in the numerator and the total energy consumption in the denominator. Due to this high dimensionality and nonlinear interdependence, obtaining a globally optimal solution in polynomial time is highly challenging.

III. PROPOSED SOLUTION

We propose a hybrid iterative approach using Block Coordinate Descent (BCD) to decompose the problem into two stages. The overall proposed architecture is illustrated in Fig. 2. In the first stage, user associations, UAV positions and IRS phase-shift coefficients are fixed, and UAV beamforming matrices are determined. We employ ZF precoding to analytically suppress inter-vehicle interference for a given network state, providing a tractable closed-form beamforming solution. In the second stage, the GNN-enhanced MADRL framework (under CTDE) jointly optimizes the high-dimensional, non-convex variables (UAV positions, IRS phase-shift coefficients, and user association indicators) to maximize global energy efficiency. These stages coordinate iteratively: the MADRL agents shape the effective channel environment, while the ZF module provides a stable, interference-free rate calculation that serves as the reward signal. This cycle repeats until convergence, ensuring a balance between analytical interference management and learned global optimization.

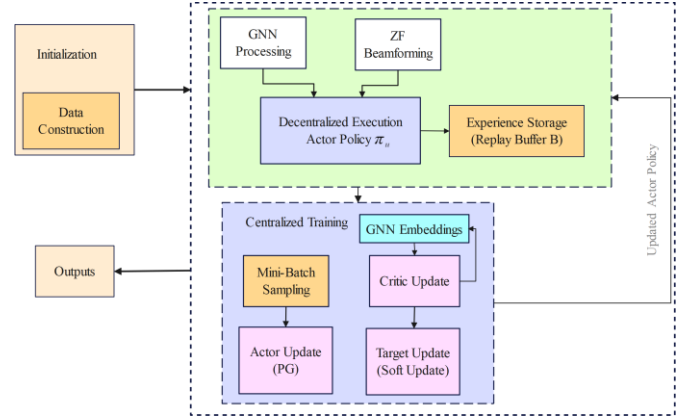


Fig.2. The workflow of the proposed approach

A. Beamforming Design

The first subproblem, P2, focuses on optimizing the transmit beamforming matrices for all UAVs, given fixed UAV positions and IRS phase-shift coefficients. The P2 formulation is given as:

$$\begin{aligned} \mathcal{P2} \quad & \max_{\mathbf{W}} \quad \eta_{ee} \\ \text{s.t.} \quad & \text{C1: } \|\mathbf{W}_u\|_F^2 \leq P, \forall u \in \mathcal{U}. \end{aligned}$$

The goal of this subproblem is to maximize the sum rate of the served vehicles while strictly suppressing inter-vehicle interference. To this end, ZF precoding is adopted because UAVs operate as multi-antenna access points in a dense vehicular environment where strong co-channel interference is the dominant limiting factor. ZF exploits the spatial degrees of freedom of the UAV antenna arrays to null inter-vehicle interference in closed form, making it a widely used and analytically tractable choice for UAV-assisted and IRS-aided networks [32].

Each UAV u serves its associated set of vehicles using a pre-coding matrix $\mathbf{W}_u$, where the ZF constraint is formulated as $\mathbf{H}_u \mathbf{W}_u = \mathbf{I}$, with $\mathbf{H}_u$ denoting the effective channel matrix between UAV u and its associated vehicles. The ZF precoder for each UAV is obtained according to the following expression:

$$\tilde{\mathbf{W}}_u = \mathbf{H}_u^\dagger = \mathbf{H}_u^H (\mathbf{H}_u \mathbf{H}_u^H)^{-1} \quad (10)$$

In (10), $\mathbf{H}_u^\dagger$ denotes the pseudo-inverse of the channel matrix $\mathbf{H}_u$. Computing the precoding matrix in this manner guarantees that inter-user interference is eliminated while maximizing the achievable sum-rate for each UAV. In cases where $\mathbf{H}_u$ is ill-conditioned (e.g., due to a low rank), the regularized zero-forcing (RZF) approach [33] can be employed:

$$\tilde{\mathbf{W}}_u = \mathbf{H}_u^H (\mathbf{H}_u \mathbf{H}_u^H + \mu \mathbf{I})^{-1} \quad (11)$$

where μ is a regularization parameter that enhances the stability of the matrix inversion.

As commonly adopted in the UAV-assisted and IRS-aided communication literature [17], this work assumes perfect channel state information (CSI) in order to focus on the joint optimization of UAV positioning, IRS configuration, and user association under the proposed CTDE-based MARL

framework. Nevertheless, the adopted RZF structure inherently improves robustness against channel estimation errors.

To satisfy constraint C1, the matrix $\widetilde{\mathbf{W}}_u$ is normalized as follows:

$$\mathbf{W}_u = \frac{\sqrt{P}}{\|\widetilde{\mathbf{W}}_u\|_F} \widetilde{\mathbf{W}}_u \quad (12)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. It should be noted that, in the initial iteration, an initial user-association strategy ($\alpha_{u,n}^{(0)}$) is required. Specifically, it is assumed that each user is initially served by the UAV providing the strongest channel gain while having the largest remaining serving capacity. This is expressed as

$$\alpha_{u,n}^{(0)} = \begin{cases} 1, & \text{if } u = \arg \max \left(\left(1 - \frac{\sum_{n=1}^N \alpha_{u,n}^{(0)}}{N_u} \right) h_{u,n}^c \right) \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

It should be noted that the candidate UAV set for each user is restricted to those UAVs satisfying the coverage condition defined in Section II, and the association rule in (14) is applied only within this feasible set.

B. Joint Optimization of User Association, UAV Positions, and IRS Phase Shifts

After determining the UAV beamforming matrices, the next step involves jointly optimizing the user association, UAV positions, and IRS phase-shift coefficients with the objective of maximizing the long-term energy efficiency in subproblem P3:

$$\begin{aligned} \mathcal{P3} \quad & \max_{\mathbf{A}, \boldsymbol{\theta}, \Phi} \quad \eta_{ee} \\ \text{s.t.} \quad & \text{C2-C5.} \end{aligned}$$

To solve subproblem P3 we adopt a learning-based approach that combines GNN feature extraction with MADRL under the CTDE paradigm. This enables scalable joint optimization in dynamic, large-scale vehicular networks.

1) *Graph Representation Using GNN*: The GNN is a message passing neural network (MPNN) that performs graph convolutions to update node embeddings based on neighboring information. This enables permutation-invariant and relationship-aware feature extraction, which is essential for capturing the dynamic inter-agent couplings in MADRL. Unlike CNNs, which rely on grid-like data structures, or multilayer perceptron (MLPs), which ignore structural relationships, GNNs directly model the non-Euclidean graph topology of the UAV-IoV system, allowing the policy to learn coordinated, context-dependent actions (e.g., repositioning a UAV based on its neighbors' positions and load) that improve overall network performance. At each time step, nodes exchange information via edges (channel coefficients), and update their representations using a graph convolution operation. The output does not depend on the order of nodes, which is crucial for dynamic multi-agent systems where agent identities change.

First, a graph representation of the network is constructed, where the nodes consist of UAVs, IRS arrays, and users (vehicles), and the edges represent the channel coefficients between UAVs, IRSs, and vehicles. The node features include UAV positions, user positions, IRS phase-shift coefficients,

and association parameters. In this stage, a GNN extracts both structural and temporal features, which are then utilized to model the complex interactions between UAVs and users [34]. The GNN is defined as

$$G_u[t] = \text{GNN}(\mathcal{G}[t]) \quad (14)$$

where $\mathcal{G}[t]$ denotes the graph at time slot t , and $G_u[t]$ encodes the neighboring UAVs, vehicles, and IRS arrays into a compact vector representation. This output subsequently supports the policy update process in the CTDE framework.

Within each GNN layer, node embeddings are iteratively updated through a neighborhood aggregation process that incorporates information from adjacent nodes. Specifically, the embedding of node i at layer $l+1$ is computed by combining its previous representation with the aggregated features of its neighboring nodes as

$$h_i^{(l+1)} = \sigma(\mathcal{W}_s h_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathcal{W}_n h_j^{(l)}) \quad (15)$$

where $h_i^{(l)}$ denotes the embedding of node i at layer l , $\mathcal{N}(i)$ represents the set of neighboring nodes, $\mathcal{W}_s$ and $\mathcal{W}_n$ are learnable weight matrices, and $\sigma(\cdot)$ is a nonlinear activation function. Through this message passing mechanism, each node embedding progressively captures both local attributes and the relational structure of the UAV-IRS-vehicle communication network.

Furthermore, due to the mobility of UAVs and vehicles, the underlying network topology evolves continuously over time. To reflect these dynamics, the heterogeneous graph representing UAVs, IRS panels, and vehicles is reconstructed at every environment time step according to the updated spatial configuration and communication links. This dynamic graph reconstruction ensures that the GNN embeddings accurately capture the current network topology and channel-dependent interactions, enabling the reinforcement learning agents to make decisions based on the latest system state.

The GNN and the Actor-Critic networks are trained jointly, without any separate pre-training phase. The GNN processes the graph representation of the system, comprising UAVs, IRS arrays, and users, and outputs node embeddings that capture structural and temporal features. These embeddings serve as the primary input to the actor network of each UAV, enabling localized decision-making (e.g., repositioning, phase shift tuning, and user association). For the centralized critic, the embeddings of all UAVs are concatenated with their corresponding action vectors to form a global state-action representation, allowing the critic to evaluate the joint impact of all agents' actions. Since the GNN is embedded in the policy and value networks, its outputs directly influence both the actor decisions and the critic evaluation. Consequently, the RL loss is backpropagated through the actor-critic networks to the GNN via standard gradient-based optimization, allowing the GNN parameters to be updated jointly with the actor and critic in an integrated manner.

2) *MARL with CTDE*: To solve Problem P3, we propose a feature-extraction-based algorithm that integrates the GNN with a CTDE reinforcement learning paradigm. This approach replaces the standard MADDPG algorithm and provides advantages such as enhanced performance, improved stability, and faster convergence [35]. Block diagram of the proposed GNN-CTDE architecture is shown in Fig. 3. The diagram illustrates the two main phases: centralized training,

dominating the reward. The cumulative local reward over a horizon T is:

$$r_u[t] = \sum_{t=1}^T \gamma[t] r_u^{\text{ins}}[t] \quad (20)$$

- **Experience Storage:** The tuple $(s[t], a[t], r^{\text{tot}}, s[t+1])$ is stored in the replay buffer.
- **Centralized Critic Update:** In this architecture, the centralized critic processes the concatenated global feature set, comprising both the GNN-produced embeddings and the actions of all UAVs, through a multi-layer neural network to predict the scalar Q-value. This structure allows the critic to capture complex interactions between UAVs and the broader environment, providing accurate gradient feedback for updating the decentralized actor networks. This processed representation is then used within the training loop, where a mini-batch of size B is sampled from the replay buffer $\mathcal{B}$ to compute the target value as:

$$y_i = r_i^{\text{tot}} + \gamma Q_{\omega_c}(s_i[t+1], a'_i[t+1]; \omega'_c) \quad (21)$$

where critic network parameters ω_c are updated by minimizing the following loss function:

$$\mathcal{L}_c(\omega_c) = \frac{1}{B} \sum_{i=1}^B (Q_{\omega_c}(s_i[t], a_i[t]; \omega_c) - y_i)^2 \quad (22)$$

and the gradient descent is updated as

$$\omega_c \leftarrow \omega_c - \mu_c \nabla_{\omega_c} \mathcal{L}_c(\omega_c) \quad (23)$$

with μ_c representing the learning rate for critic.

- **Decentralized Actor Updates:** The centralized critic computes the policy gradients for each agent, and the actor parameters ω_u are updated as:

$$\nabla_{\omega_u} J \quad (24)$$

$$\approx \frac{1}{B} \sum_{i=1}^B \nabla_{a_u} Q_{\omega_c}(s_i[t], a_i[t]; \omega_c) \cdot \nabla_{\omega_u} \pi_u(s_{u,i}[t])$$

where J represents the long-term expected reward of the proposed MARL framework which is mathematically defined as:

$$J = \mathbb{E} \left[\sum_{t=0}^T \gamma[t] r_u^{\text{ins}}[t] \right] \quad (25)$$

and the gradient descent is updated as

$$\omega_u \leftarrow \omega_u - \mu_a \nabla_{\omega_u} J \quad (26)$$

with μ_a representing the learning rate for actors.

To improve training stability, separate copies of the actor and critic networks, called target networks, are maintained. These networks ($Q_{\omega_{c'}}$ for the critic and $\pi_{u'}$ for each actor) are used to compute target values instead of the rapidly changing main networks. The target networks are not updated every gradient step; instead, they are adjusted gradually to follow the main networks, a process known as soft update. The parameters of the target critic network $\omega_{c'}$ and target actor network $\omega_{u'}$ are updated gradually to match their respective main networks according to:

$$\begin{aligned} \omega_{c'} &\leftarrow \tau \omega_c + (1 - \tau) \omega_{c'} \\ \omega_{u'} &\leftarrow \tau \omega_u + (1 - \tau) \omega_{u'} \end{aligned} \quad (27)$$

where $0 < \tau \ll 1$ is the soft-update factor.

After training converges, the system transitions to the execution phase. Each UAV independently selects its action according to its trained local policy π_u using only local observations:

$$a_u[t] = \pi_u(s_u[t]) \quad (28)$$

This phase is fully decentralized and requires no global information or centralized critic.

Algorithm 1: Proposed GNN-CTDE approach for solving problem $\mathcal{P}1$.

Input: Training Dataset $\mathcal{D}$, System Parameters $\lambda_1 - \lambda_5$, N_u , P , $\delta_{\max}$, $\delta_{\min}$, Number of Episodes E , Time Horizon T , Neural Networks for Actor π_u and Critic Q_{ω_c} , Discount Factor γ , Soft update factor τ , Learning Rates μ_a, μ_c .

Output: $\mathbf{A}, \mathbf{W}_u, \Theta_u, \Phi_u^{\text{UAV}}$.

1. Initialize UAV positions $\Phi_u^{(0)}$, IRS phase shifts $\Theta_u^{(0)}$, and user associations $\mathbf{A}^{(0)}$, π_u and Q_{ω_c} , replay buffer $\mathcal{B}$.
 2. **for each episode** $e = 1, \dots, E$
 3. **for each time step** $t = 1, \dots, T$
 4. Fetch current environment state from $\mathcal{D}$ and construct graph $\mathcal{G}[t]$.
 5. Compute node embeddings $G_u[t]$ using GNN.
 6. Compute ZF beamforming matrices $\mathbf{W}_u[t]$ for each UAV by Eq. (12).
 7. **// Decentralized action selection:**
 8. Each UAV selects action $a_u[t]$ using actor policy π_u .
 9. Each UAV Executes action $a_u[t]$, observes new state $s_u[t+1]$, and rewards $r_u[t]$.
 10. Compute $r^{\text{tot}} = \sum_{u=1}^U r_u^{\text{ins}}[t]$ from Eq. (18).
 11. Store experience $(s[t], a[t], r^{\text{tot}}, s[t+1])$ in replay buffer $\mathcal{B}$.
 12. **end for**
 13. **// Centralized Training:**
 14. Sample mini-batch from $\mathcal{B}$.
 15. Update critic Q_{ω_c} by Eq. (21).
 16. Update actor π_u by Eq. (23).
 17. Soft update target networks $Q_{\omega_{c'}}$ and $\pi_{u'}$ by Eq. (26).
 18. **end for**
 19. **return** $\pi_u, \mathbf{A}, \mathbf{W}_u, \Theta_u, \Phi_u^{\text{UAV}}$.
-

C. Proposed Algorithm

Algorithm 1 outlines the main steps of the proposed method for solving Problem $\mathcal{P}1$. The process begins with data construction and initialization. Here, *dataset construction* refers to the dynamic generation of environment data during training, rather than relying on a fully pre-collected static dataset. The dataset is dynamically updated at each time step based on the current positions of UAVs, vehicles, and IRS panels, as well as the corresponding LoS/NLoS channel states and user associations. This enables the learning process to adapt to time-varying network conditions and mobility patterns in real time. In parallel, ZF beamforming is computed for the multi-user downlink communication model. For each UAV, the policy network π_u uses the extracted GNN features along with channel state information to make decisions regarding UAV repositioning, IRS phase-shift tuning, and user association. The selected actions, observed states, and rewards, are stored in the centralized replay buffer. Since the replay buffer is centralized, all UAV agents should

periodically transmit their local state, action, reward, and next-state tuples to the central learner during the training. This creates additional signaling overhead, requiring reliable control links and sufficient bandwidth. However, because centralized training is conducted in a controlled environment where strict real-time service demands are absent, this overhead is tolerable. In the decentralized execution phase, such signaling is eliminated, allowing each UAV to operate autonomously without incurring extra communication cost. During centralized training, experience tuples are sampled in mini-batches from the replay buffer. The actor networks are updated using policy gradients, while the critic network is updated via temporal-difference (TD) learning. Target networks are softly updated to maintain training stability. The updated policies are then deployed in the decentralized execution environment, where each agent operates independently and optimally without requiring global information or centralized coordination.

IV. PERFORMANCE EVALUATION

In this section, the performance of the proposed algorithm is evaluated through the simulations. Specifically, we report key performance metrics including energy efficiency and convergence speed. All performance curves are obtained by averaging over 100 independent Monte-Carlo realizations. All simulations were conducted on a high-performance computing system equipped with an NVIDIA TITAN RTX graphics card featuring 24 GB of VRAM, a multi-core CPU, and 200 GB of RAM. The software environment was configured using Python 3.10 with PyTorch v2.0 and the PyTorch Geometric library. GPU acceleration was enabled via CUDA v11.6, and the Deep Graph Library (DGL) v1.1.2 was employed for efficient graph-based computations.

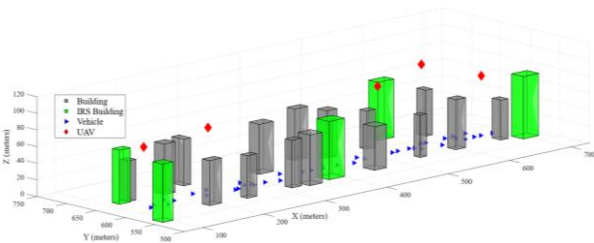


Fig 4. 3D view of the system model

A. Environment Modeling

As illustrated in Fig. 4, the system under study is a vehicular network supported by multiple UAVs, each paired with a dedicated IRS panel installed on the tallest building within its coverage zone. Multiple vehicles move dynamically along two parallel highways, and tall buildings line both sides of the highways. To evaluate the performance of the UAV- and IRS-aided vehicular communications, a comprehensive dataset was built to model the wireless channels among UAVs, vehicles, and IRS panels. This dataset combines realistic mobility and environmental data and supports both LoS and NLoS channel conditions. The modeled environment includes four main components: Vehicle trajectories from the NGSIM US-101 dataset are extracted for a $700\text{ m} \times 250\text{ m}$ highway segment and synchronized with UAV-IRS simulation times. Buildings are defined in 3D, some equipped with IRS panels on each wall. Initial UAV positions are obtained through K-Means clustering of vehicle locations to

cover dense traffic zones. Line-of-sight conditions for UAV-vehicle and vehicle-IRS links are determined via geometric obstruction checks using building data [18], [38].

B. Graph-Based Environment Modeling

The simulation environment is represented as a heterogeneous graph with UAVs, vehicles, and IRS panels as nodes, and bidirectional edges encoding their communication links. A custom heterogeneous GNN embeds these nodes into feature vectors. Raw node features are first processed by a linear transformation layer whose output size depended on the node type, ranging from 1 for minimal-attribute nodes (i.e., IRS panels) to 64 for rich-information nodes (i.e., UAVs). The transformed features are aggregated via a GraphSAGE layer to capture local and structural connectivity, and then mapped through output layers (64–256 units) based on node type. The final embeddings are fed into the agent networks, i.e., the actor and critic neural networks of each UAV, for decision-making in the MARL framework.

C. Actor and Critic Network Architecture

Each UAV is equipped with an independent actor network that maps its 256-dimensional embedding to an action. Specifically, the actor network takes as input a 256-dimensional feature vector obtained from the GNN and produces an output composed of three components: the UAV repositioning vector $(\Delta x_u, \Delta y_u)$, the IRS phase-shift parameters θ_k , and the user-association parameters $\alpha_{u,n}$ for each UAV. The actor network consists of three fully-connected hidden layers with 128, 256, and 256 neurons, respectively, with ReLU activation functions applied after each layer to introduce nonlinearity.

The centralized critic is implemented as a fully connected multi-layer neural network with four hidden layers of 512, 512, 256, and 128 neurons, respectively. Each hidden layer is followed by a ReLU activation function, and the final output layer is a single-neuron linear layer that produces the scalar Q-value. The hyperparameters used during training, along with the simulation environment details, are summarized in Table II. Throughout this section, the term energy efficiency refers to η_{ee} as defined in Problem $\mathcal{P}1$.

D. Baselines for Comparison and Evaluation

To comprehensively evaluate the proposed method from multiple perspectives, several baseline schemes were implemented and incorporated into the simulation results for comparison. The first baseline, referred to as GF-MRT (Greedy user allocation with Fixed IRS phases and Maximum Ratio Transmission beamforming), assigns each user to the nearest UAV. The IRS phase-shift configuration remains fixed, and MRT beamforming is applied, with the total transmit power equally distributed among the users. This scheme highlights the ability of the proposed method to adaptively control interference and expand the effective spectral efficiency region. The second baseline, RP-ZF (Random UAV Placement with ZF beamforming), randomly selects UAV positions and applies ZF beamforming. This configuration serves to demonstrate the importance of learning UAV positions and performing dynamic UAV deployment optimization. The third baseline, DDPG-NGNN (Centralized DDPG without GNN), uses a single-agent DDPG to control

the entire system in a centralized manner, without employing any graph structure or message passing among agents. The comparison with this method demonstrates the superiority of the proposed GNN-CTDE approach in terms of scalability and distributed adaptability. The fourth baseline, DP-FP (Density-based UAV Positioning with Fixed IRS phases), positions UAVs according to the areas of highest vehicular density while keeping IRS phase shifts fixed. The objective of comparing against this scheme is to assess the benefits of multi-agent learning and IRS phase optimization. Finally, the fifth baseline, MADDPG-NGNN, employs the same multi-agent architecture as MADDPG but omits graph embeddings, feeding the actor and critic networks directly with raw vehicle and UAV state information. This comparison is intended to quantify the performance impact of incorporating graph-based learning into the multi-agent deep reinforcement learning framework.

Table II. Parameter Settings

Parameter	Value
Carrier Frequency	2GHz
Number of UAVs (training & test) (U)	5
Number of UAV Antenna Arrays (M)	10
IRS Elements per UAV (K)	8
UAV Altitude (h_u)	100m
IRS Panel Altitude (h_r)	50-70m
Vehicle Count (N)	50
UAV Maximum Flight Distance (δ_{max})	60m
Maximum Vehicles per UAV (N_u)	10
Noise Power Spectral Density	-169dBm/Hz
Pathloss at Reference Distance (ρ)	-30dB
Pathloss Exponent (ζ)	2.2
GNN Embedding Dimension	256
Replay Buffer Capacity	100,000
Learning Rate μ_a (Actor)	3e-4
Learning Rate μ_c (Critic)	2e-3
Discount Factor (γ)	0.95
Soft Update Factor (τ)	0.01
Optimizer	Adam
Batch Size (B)	128
Episodes (E)	1500
Penalty Coefficients ($\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$)	0.001, 1, 0.1, 0.5, 0.5
Timesteps per Episode (T) (training/test)	100/20
Timestep Duration (Δt)	10 sec

E. Simulation Results

Fig. 5 depicts the variation of the system energy efficiency during the training phase as a function of the number of training episodes, for two different batch sizes of 64 and 128. This figure illustrates both the stability and the learning speed of the proposed multi-agent algorithm throughout the training process. As observed, the proposed method approaches a stable operating region after approximately 400 episodes, beyond which the fluctuations in energy efficiency diminish and the algorithm converges to a relatively optimal policy. When the batch size is 128, the convergence of energy efficiency occurs more rapidly than in the case of a batch size of 64. This is because a larger batch size enables the model to process a greater number of data features simultaneously, thereby achieving more effective and faster learning. Nevertheless, due to the incorporation of a graph-based structure and the CTDE framework, the proposed algorithm is able to attain satisfactory convergence even under smaller batch sizes.

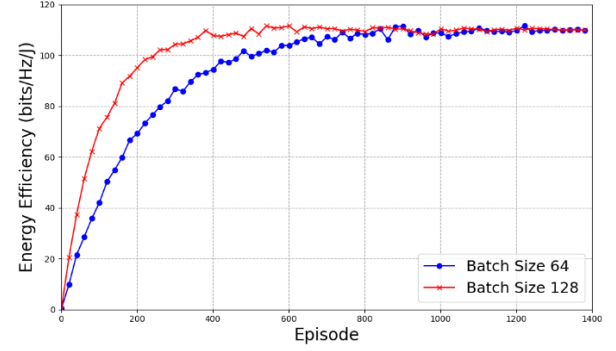


Fig. 5. Energy efficiency convergence of the proposed multi-agent algorithm during training

Fig. 6 illustrates the variation of the system energy efficiency with respect to the UAV transmit-power budget. To provide additional statistical insight, the error bars around the curves represent the variability range obtained across multiple independent runs, confirming the stability of the proposed method under different realizations. The variability slightly increases at higher UAV transmit powers due to the stronger impact of channel fluctuations and interference dynamics in high-power operating regimes. Moreover, the curve initially exhibits an increasing trend, followed by a gradual slowdown that leads to saturation. This behavior can be attributed to the fact that, at low power budgets, increasing transmit power improves the data rate and consequently the energy efficiency. In this regime, the total energy consumption remains relatively low, while the higher power improves the SINR, enhances transmission rates, and reduces the need for frequent UAV repositioning, thereby further conserving energy. However, beyond a certain threshold, further increases in power lead to a disproportionate rise in energy consumption, while the data rate experiences only marginal improvement due to channel limitations, interference, and link-capacity saturation. In this region, the ratio of data rate to energy consumption, i.e., the energy efficiency, begins to decline. The proposed GNN-CTDE method, using ZF beamforming, maintains high data rates even at elevated powers, shifting the efficiency peak to higher budgets compared with MRT-based GF-MRT and non-optimized RP-ZF schemes. Methods lacking graph-based learning or joint UAV/IRS optimization (DDPG-NGNN, DP-FP, MADDPG-NGNN) exhibit weaker scalability, coordination, or interference suppression, leading to earlier saturation and lower efficiency in high-power regimes. The proposed GNN-CTDE method achieves an average energy-efficiency improvement of approximately 7.9% over the best competing baseline.

Fig. 7 illustrates the variation of system energy efficiency with respect to the number of vehicular users, which also serves as a scalability evaluation under increasing network size. As the number of vehicles increases, the energy efficiency initially improves due to enhanced user diversity, while excessive network loading eventually leads to performance saturation. Importantly, the proposed GNN-CTDE framework exhibits a significantly slower degradation trend compared to baseline methods as the network size grows, demonstrating its scalability and robustness in large-scale vehicular scenarios. It achieves an average energy-efficiency improvement of approximately 7.8% compared with the best competing baseline across different numbers of vehicles. Owing to the graph-structured state representation and

parameter sharing among agents, the proposed framework avoids exponential growth in the joint state-action space and maintains stable performance trends as the network scale increases. This property is particularly important for dense vehicular environments, where centralized or non-graph-based MARL methods often suffer from rapid performance degradation.

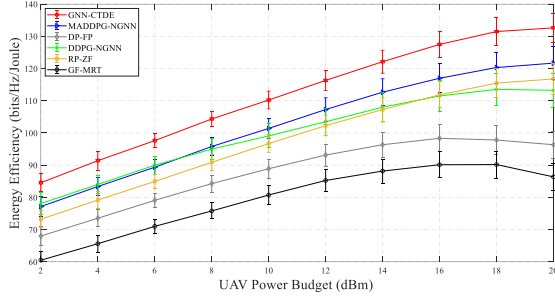


Fig. 6. System energy efficiency versus UAV transmit-power budget.

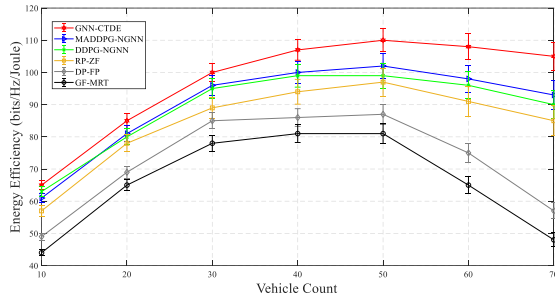


Fig. 7. System energy efficiency versus vehicle count.

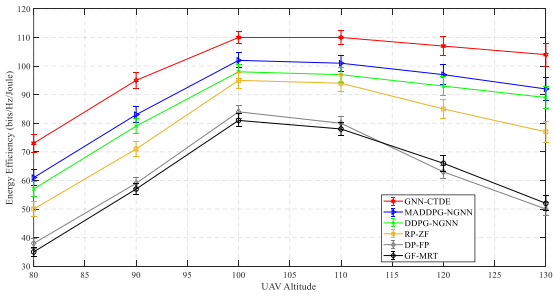


Fig. 8. System energy efficiency versus UAV altitude.

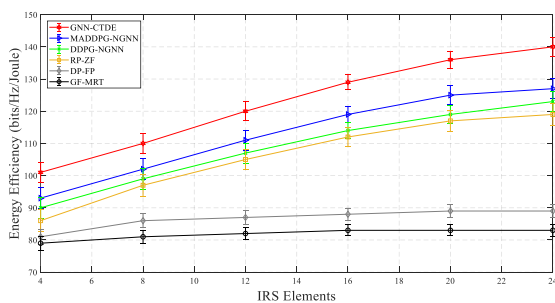


Fig. 9. System energy efficiency versus the number of IRS elements.

Fig. 8 presents the system energy efficiency as a function of UAV altitude. At lower heights, increasing altitude enhances

the probability of LoS links and expands vehicle coverage, thereby improving energy efficiency. However, beyond an optimal altitude, further increases extend the propagation distance and attenuate the signal, leading to reduced data rates and a gradual decline in energy efficiency. This indicates the presence of an optimal height that balances coverage, signal strength, and energy consumption. In the proposed GNN-CTDE scheme, intelligent horizontal positioning of UAVs, combined with ZF beamforming and multi-agent learning, yields a higher peak energy efficiency at the optimal altitude than the reference methods. In Fig. 8, the proposed GNN-CTDE framework achieves an average energy-efficiency improvement of approximately 12.4% over the best competing baseline.

Fig. 9 depicts the variation of energy efficiency with the number of IRS elements per panel. Larger IRS arrays provide finer and more targeted control over reflected paths, enhancing UAV-IRS-vehicle channel quality and increasing data rates. In the proposed GNN-CTDE scheme, IRS phase shifts are jointly optimized with UAV positioning and user association, enabling the algorithm to fully exploit the additional reflecting elements. This leads to sustained energy-efficiency gains as long as the data rate continues to grow, while also reducing UAV repositioning energy costs. Beyond a certain array size, however, channel and UAV power constraints cause data-rate saturation, and the energy-efficiency improvement tapers off. Methods without IRS phase optimization (e.g., DP-FP, GF-MRT) or without graph-based agent coordination (e.g., MADDPG-NGNN) benefit far less from increased IRS size. Overall, the proposed algorithm makes more effective use of large-scale IRS arrays, maintaining superior energy efficiency at higher scales and achieving an average energy-efficiency improvement of approximately 8.7% over the best competing baseline.

Fig. 10 illustrates the 3D flight trajectories of UAVs over five consecutive time slots during the testing phase. After training, the UAVs employ the learned policies to determine their optimal positions at each time step. As shown, they gradually move toward energy-efficient regions by accounting for vehicular user distribution and IRS panel locations, ultimately stabilizing at positions that maximize system performance. This demonstrates the capability of the proposed model to learn effective UAV deployment strategies and maintain stable behavior in unseen scenarios.

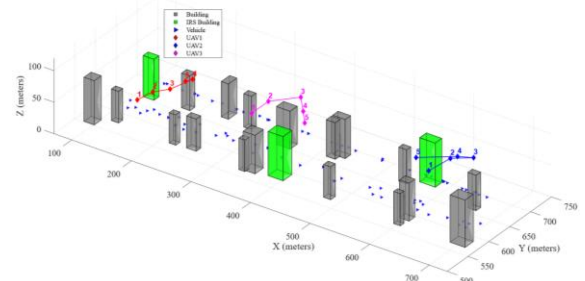


Fig. 10. UAV flight trajectories in the 2D plane during the testing phase.

F. Sensitivity Analysis

To evaluate the robustness of the proposed reward design, we conduct a sensitivity analysis on two key coefficients, namely λ_1 and λ_3 . These parameters are selected because they

directly influence two critical components of the learning process: UAV mobility control and IRS phase constraint enforcement. Specifically, λ_1 regulates the movement penalty of the UAVs and thus affects trajectory exploration and stability, while λ_3 controls the constraint on the IRS phase shifts, impacting feasibility and convergence behavior. Since these two terms strongly shape the policy update dynamics, they are the most influential coefficients in the reward structure.

Fig. 11. (a) illustrates the convergence of the average cumulative local reward for different values of $\lambda_1 \in \{0.0005, 0.001, 0.005\}$. When $\lambda_1 = 0.0005$, the mobility penalty is reduced, encouraging more aggressive exploration and resulting in faster initial convergence with slightly larger fluctuations. Conversely, increasing the coefficient to $\lambda_1 = 0.005$ enforces conservative UAV movements, which slows down learning and yields a lower final reward. The selected value $\lambda_1 = 0.001$ achieves a proper balance between exploration and stability, leading to smooth convergence and superior overall performance.

Similarly, Fig. 11. (b) presents the convergence behavior for different values of $\lambda_3 \in \{0.05, 0.1, 0.2\}$. A smaller value ($\lambda_3 = 0.05$) weakens the IRS phase constraint, allowing greater flexibility in action updates and thus faster exploration, albeit with increased oscillations. In contrast, a larger value ($\lambda_3 = 0.2$) imposes stricter penalties on infeasible phase adjustments, which slows convergence and slightly reduces the final reward. The intermediate choice $\lambda_3 = 0.1$ provides the best trade-off between constraint enforcement and learning flexibility, ensuring stable and efficient convergence.

Overall, the results demonstrate that the proposed DRL framework maintains consistent convergence behavior under moderate variations of these reward coefficients, confirming that the selected parameter values are well justified and that the algorithm is robust to practical tuning deviations.

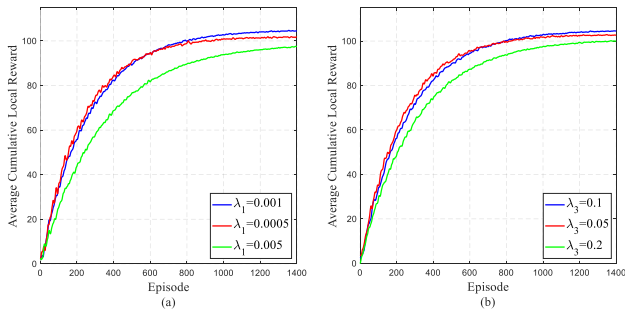


Fig. 11. Convergence behavior of the average cumulative local reward for different values of the reward penalty coefficients: (a) λ_1 and (b) λ_3 .

V. DISCUSSION ON LIMITATIONS

While the proposed GNN-MADRL framework for UAV-assisted IRS achieves significant performance gains in terms of sum-rate maximization and efficient resource management, it is essential to acknowledge several inherent limitations and practical considerations that warrant further investigation.

- **Imperfect CSI and Robustness:** Channel uncertainties may reduce precoding effectiveness and bias the learned policies. Although RZF provides a baseline for

robustness, integrating stochastic or partially observable CSI models and adaptive learning remains a key direction to enhance the practical performance of the framework.

- **Trade-off of Problem Decomposition:** The proposed two-stage design improves tractability and real-time deployability, but fixing the ZF beamformer may limit global optimality compared with fully joint optimization, which would involve a much larger action-state space and slower convergence.
- **Representativeness beyond US-101:** The evaluation on the NGSIM US 101 dataset provides a realistic highway baseline, but broader validation is needed for complex urban and intersection scenarios. The proposed CTDE-GNN framework can be extended to such topologies by updating graph adjacency with 3D building and LoS/NLoS information, which remains an important direction for future study.
- **Integration with Mobility-Aware Routing and Logistics Frameworks:** This work focuses on communication-layer optimization and does not explicitly incorporate vehicular routing or logistics decisions. Future research could integrate mobility-aware routing strategies, such as traffic-light-aware vehicular routing [39], or logistics-aware routing and task-assignment frameworks for multi-truck multi-drone systems [40], [41], to enable cross-layer communication-transportation co-design.

VI. COMPUTATIONAL COMPLEXITY AND SIGNALING OVERHEAD

The computational complexity of the proposed GNN-CTDE algorithm is dominated by three components: (i) feature-vector extraction via the GNN, (ii) ZF beamforming matrix computation, and (iii) actor-critic network training. In our system model, the total number of graph edges is approximated as $N_E \approx 2UN$, accounting for UAV-user and user-IRS interactions, where the number of vehicles N is much larger than the number of UAVs U . Let d denote the GNN embedding dimension, which scales with the number of antennas and IRS elements, i.e., $d \propto (M + K + c)$, where c counts scalar system features (e.g., position, velocity, and power metrics). Accordingly, the GNN feature extraction incurs a complexity of $\mathcal{O}(N_E d^2) \approx \mathcal{O}(2UN(M + K + c)^2)$. The ZF beamforming requires inversion of the channel matrix, whose dimensions depend on the number of users N_u , leading to complexity of $\mathcal{O}(UN_u^3)$. Consequently, when the number of users per UAV remains bounded, the overall execution time increases approximately linearly with U . This linear growth contrasts with centralized optimization frameworks, where computation time often grows super-linearly or even exponentially with the number of coordinated agents. Therefore, although larger UAV fleets naturally incur higher processing time, the proposed GNN-MADRL framework preserves tractable timing performance for moderately scaled deployments.

During centralized training, actor and critic networks are updated using mini-batch stochastic gradient descent with complexity $\mathcal{O}(BD_{NN})$, where B and D_{NN} denote the mini-batch size and the number of trainable parameters, respectively. Considering E training episodes with T time steps per episode, the overall training-phase complexity is given by

$$\mathcal{O}(ET[2UN(M + K + c)^2 + UN_u^3 + BD_{NN}]). \quad (29)$$

During centralized training, signaling overhead arises from sharing local states and link information to construct the global graph. The per-step communication load scales as

$$\mathcal{O}(U(M + K + c) + c_n N + 2UN). \quad (30)$$

which can be simplified to $\mathcal{O}(UK + UN)$. This overhead occurs entirely offline during CTDE and does not affect the operational bandwidth of the vehicular network.

In the decentralized execution phase, the signaling volume per control step depends primarily on the number of active vehicles and UAVs. Let n_f the number of locally shared features or control parameters per agent, and Δt the control interval. Assuming 32-bit floating-point representation for each transmitted quantity, the required signaling rate can be expressed as

$$BW_{sig} = \frac{32(V + U) n_f}{\Delta t} [\text{bit/s}] \quad (31)$$

Since n_f is fixed by the proposed state/action design and Δt is on the order of seconds, the resulting BW_{sig} remains in the order of a few kilobits per second even for large fleets, confirming its negligible impact on system bandwidth.

Compared with global MINLP-based solutions, whose complexity grows exponentially with problem size and require exchanging full channel state information at each iteration, and fully centralized DDPG schemes with extremely large state-action spaces, the proposed framework achieves significantly lower computational and signaling costs while preserving scalability and real-time feasibility.

Table III summarizes the dominant computational and signaling complexity orders of the proposed method and representative baselines.

Table III. Comparison of Computational Complexity and Signaling Overhead

Method	Complexity order	Signaling overhead
Proposed	$\mathcal{O}(ET[2UNK^2 + UN_u^3])$	$\mathcal{O}(UK + c_n N + UN)$
Global MINLP	Exponential in the total number of variables	$\mathcal{O}(UNK)$
Centralized DDPG	$\mathcal{O}(ET[UN_u^3 + B\bar{D}_{NN}])$, $\bar{D}_{NN} \gg D_{NN}$	$\mathcal{O}(UK + \bar{c}_n N + UN)$, $\bar{c}_n \gg c_n$

In addition to asymptotic complexity analysis, we further compare the practical computational overhead of the proposed framework with a MARL baseline without GNN integration in terms of average algorithmic training and execution time as provided in Table IV. Although the proposed GNN-enhanced MADRL incurs a moderate increase in per-episode training time due to graph message passing and embedding computation, Fig. 5 demonstrates that the proposed method converges to a stable operating region after approximately 400 training episodes (for the batch size of 128). Therefore, despite the slightly higher per-episode computational overhead, the total training duration required to reach convergence remains practical. Moreover, during decentralized execution, the inference-time overhead is small and suitable for real-time deployment. The reported inference time excludes environment simulation and channel generation.

Table IV. Average Algorithmic Training and Inference Time

Method	Training time per episode	Inference time per slot
MADRL (without GNN)	8.2 ms	1.1 ms
Proposed GNN-MADRL	11.5 ms	1.4 ms

VII. CONCLUSION

This paper presented an intelligent and scalable framework for improving energy efficiency in UAV-assisted IoV networks. The proposed architecture integrates multi-antenna UAVs, IRS-assisted propagation control, and graph-structured multi-agent reinforcement learning to jointly optimize UAV deployment, IRS phase configuration, and user association. By combining ZF-based interference suppression with a GNN-enhanced CTDE learning strategy, the framework effectively captures the dynamic interactions among vehicles, UAVs, and the wireless environment. A realistic simulation platform was developed using the NGSIM traffic dataset together with 3D modeling of vehicles, buildings, and aerial nodes. Numerical results demonstrate that the proposed approach achieves improved energy-efficiency stability and favorable scalability compared with benchmark schemes. Future research may consider robust learning under imperfect CSI, more tightly coupled joint optimization beyond the current problem decomposition, validation in more complex urban traffic scenarios beyond the US-101 dataset, and integration with mobility-aware routing or logistics-aware multi-drone coordination frameworks to enable deeper cross-layer optimization in intelligent transportation systems.

REFERENCES

- [1] J. Li, J. Wu and X. Wang, "Multi-Agent DRL-Driven Age-Energy Efficiency Optimization in STAR-RIS Assisted Consumer-Grade Unmanned Vehicle Applications," in *IEEE Transactions on Consumer Electronics*, doi: 10.1109/TCE.2025.3635290.
- [2] Y. Wang, X. Li, X. Yi and S. Jin, "Joint User Scheduling and Precoding for RIS-Aided MU-MISO Systems: A MADRL Approach," in *IEEE Transactions on Communications*, vol. 73, no. 6, pp. 3880-3893, 2025, doi: 10.1109/TCOMM.2024.3496745.
- [3] H. Dutta, A. K. Bhuyan and S. Biswas, "Reinforcement learning based flow and energy management in resource-constrained wireless networks," *Computer Communications*, vol. 202, pp. 73-86, 2023, doi: 10.1016/j.comcom.2023.02.011.
- [4] H. Zhao, W. Sun, Y. Ni, W. Xia, G. Gui and C. Zhu, "Deep Deterministic Policy Gradient-Based Rate Maximization for RIS-UAV-Assisted Vehicular Communication Networks," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 15732-15744, 2024, doi: 10.1109/TITS.2024.3440391.
- [5] Q. Gao, R. Zhong, H. Shin and Y. Liu, "MARL-Based UAV Trajectory and Beamforming Optimization for ISAC System," in *IEEE Internet of Things Journal*, vol. 11, no. 24, pp. 40492-40505, 15 Dec.15, 2024, doi: 10.1109/IJOT.2024.3453195.
- [6] J. Li, Y. Yang, Z. He, H. Wu, H. Shi and W. Chen, "Cournot policy model: Rethinking centralized training in multi-agent reinforcement learning," *Information Sciences*, vol. 677, pp. 120983, 2024, doi: 10.1016/j.ins.2024.120983.
- [7] M. Ahmed *et al.*, "Efficient Resource Management for NOMA-Enabled UAV Communications in 6G IRS-Assisted Vehicular Networks," *IEEE Transactions on Intelligent Transportation Systems*, Early Access, 2025, doi: 10.1109/TITS.2025.3549224.
- [8] N. Sehito *et al.*, "Optimizing User Association, Power Control, and Beamforming for 6G Multi-IRS Multi-UAV NOMA Communications in Smart Cities," in *IEEE Transactions on Consumer Electronics*, vol. 70, no. 3, pp. 5702-5710, Aug. 2024, doi: 10.1109/TCE.2024.3388596.

- [9] P. Wang, D. Li, Y. Zhang and X. Chen, "UAV-Assisted Vehicular Communication System Optimization with Aerial Base Station and Intelligent Reflecting Surface," *IEEE Transactions on Intelligent Vehicles*, Early Access, 2023, doi: 10.1109/TIV.2023.3324385.
- [10] Y. Li and A. H. Aghvami, "Radio Resource Management for Cellular-Connected UAV: A Learning Approach," in *IEEE Transactions on Communications*, vol. 71, no. 5, pp. 2784-2800, May 2023, doi: 10.1109/TCOMM.2023.3262826.
- [11] H. Mei, K. Yang, Q. Liu and K. Wang, "3D-Trajectory and Phase-Shift Design for RIS-Assisted UAV Systems Using Deep Reinforcement Learning," in *IEEE Transactions on Vehicular Technology*, vol. 71, no. 3, pp. 3020-3029, 2022, doi: 10.1109/TVT.2022.3143839.
- [12] Y. Liu, C. Huang, G. Chen, R. Song, S. Song and P. Xiao, "Deep Learning Empowered Trajectory and Passive Beamforming Design in UAV-RIS Enabled Secure Cognitive Non-Terrestrial Networks," in *IEEE Wireless Communications Letters*, vol. 13, no. 1, pp. 188-192, Jan. 2024, doi: 10.1109/LWC.2023.3325066.
- [13] U. A. Mughal, Y. Alkhrijah, A. Almadhor and C. Yuen, "Deep Learning for Secure UAV-Assisted RIS Communication Networks," in *IEEE Internet of Things Magazine*, vol. 7, no. 2, pp. 38-44, March 2024, doi: 10.1109/IOTM.001.2300132.
- [14] A. I. Ameur, O. S. Oubbati, A. Rachedi, A. Arishi and M. Atiquzzaman, "Intelligent UAV Caching and Energy Management in 6G Networks," in *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 3175-3192, 2026, doi: 10.1109/TNSE.2025.3628171.
- [15] Z. Wang, X. Zhang, W. Ding, Y. Dong and X. Chen, "A Novel Integrated Sensing and Communication Scheme in UAVs-Enabled Vehicular Networks with MARL-Driven Adaptive Control" in *IEEE Transactions on Mobile Computing*, vol., no. 01, pp. 1-16, PrePrints 5555, doi: 10.1109/TMC.2025.3591259.
- [16] L. Eslami, G. Kianfar, J. Abouei, and A. Mohammadi, "Hierarchical resource optimization in multi-UAV RIS-assisted ISAC networks with uplink NOMA," *IEEE Internet of Things Journal*, early access, Nov. 2025.
- [17] X. Liu, Y. Liu and Y. Chen, "Machine Learning Empowered Trajectory and Passive Beamforming Design in UAV-RIS Wireless Networks," in *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 7, pp. 2042-2055, July 2021, doi: 10.1109/JSAC.2020.3041401.
- [18] L. Wu, C. Zhang, B. Zhang, J. Du and J. Qu, "Toward Energy-Efficiency: Integrating MATD3 Reinforcement Learning Method for Computational Offloading in RIS-Aided UAV-MEC Environments," in *IEEE Internet of Things Journal*, vol. 12, no. 14, pp. 26582-26595, 15 July 15, 2025, doi: 10.1109/JIOT.2025.3560835.
- [19] S. A. Zakaryia, M. Meaad, T. Nabil, et al., "Task offloading and resource allocation for multi-UAV asset edge computing with multi-agent deep reinforcement learning," *Computing*, vol. 107, p. 126, 2025, doi: 10.1007/s00607-025-01472-5.
- [20] Y. Xu, L. Zheng, X. Wu, Y. Tang, W. Liu and D. Sun, "Joint Resource Allocation for UAV-Assisted V2X Communication with Mean Field Multi-Agent Reinforcement Learning," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 1, pp. 1209-1223, Jan. 2025, doi: 10.1109/TVT.2024.3466116.
- [21] Y. Li, A. S. Madhukumar, T. Zheng Hui Ernest, G. Zheng, W. Saad, and A. Hamid Aghvami, "Energy-Efficient UAV-Driven Multi-Access Edge Computing: A Distributed Many-Agent Perspective," *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 8405-8420, Sep. 2025, doi: 10.1109/tcomm.2025.3552746.
- [22] A. A. Khalil, M. Y. Selim, and M. A. Rahman, "Deep learning-based energy harvesting with intelligent deployment of RIS-assisted UAV-CFmMIMOs," *Computer Networks*, vol. 229, p. 109784, 2023, doi: 10.1016/j.comnet.2023.109784.
- [23] Q. Xu, Q. You, Y. Gong, X. Yang and L. Wang, "RIS-Assisted UAV-Enabled Green Communications for Industrial IoT Exploiting Deep Learning," in *IEEE Internet of Things Journal*, vol. 11, no. 16, pp. 26595-26609, 15 Aug. 15, 2024, doi: 10.1109/JIOT.2024.3369687.
- [24] Q. Wang, Y. Lu, W. Chen, B. Ai, Z. Zhong and D. Niyato, "GNN-Enabled Optimization of Placement and Transmission Design for UAV Communications," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 6656-6661, April 2025, doi: 10.1109/TVT.2024.3514860.
- [25] Y. Pan, X. Wang, Z. Xu, N. Cheng, W. Xu and J. -J. Zhang, "GNN-Empowered Effective Partial Observation MARL Method for AoI Management in Multi-UAV Network," in *IEEE Internet of Things Journal*, vol. 11, no. 21, pp. 34541-34553, 1 Nov. 1, 2024, doi: 10.1109/JIOT.2024.3447774.
- [26] X. Tang et al., "Deep Graph Reinforcement Learning for UAV-Enabled Multi-User Secure Communications," in *IEEE Transactions on Mobile Computing*, vol. 24, no. 9, pp. 8780-8793, Sept. 2025, doi: 10.1109/TMC.2025.3558790.
- [27] E. Eldeeb and H. Alves, "Multi-agent meta-offline reinforcement learning for timely UAV path planning and data collection," arXiv e-prints, Jan. 2025, doi: 10.48550/arXiv.2501.
- [28] A. Mondal, D. Mishra, G. C. Alexandropoulos, A. Al-nahari and R. Jäntti, "Multiagent Reinforcement Learning for Offloading Cellular Communications with Cooperating UAVs," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 4, pp. 9344-9358, Aug. 2025, doi: 10.1109/TAES.2025.3554150.
- [29] C. Dutriez, O. S. Oubbati, C. Gueguen and A. Rachedi, "Adaptive and Scalable Cluster Head Management for mMTC using Sampling-Based Agglomerative Clustering and Localized Reclustering," *GLOBECOM 2025 - 2025 IEEE Global Communications Conference*, Taipei, Taiwan, 2025, pp. 5223-5228, doi: 10.1109/GLOBECOM59602.2025.11431923.
- [30] S. D. Muruganathan et al., "An Overview of 3GPP Release-15 Study on Enhanced LTE Support for Connected Drones," in *IEEE Communications Standards Magazine*, vol. 5, no. 4, pp. 140-146, December 2021, doi: 10.1109/MCOMSTD.0001.1900021.
- [31] X. Qin, Z. Song, T. Hou, W. Yu, J. Wang and X. Sun, "Joint Optimization of Resource Allocation, Phase Shift, and UAV Trajectory for Energy-Efficient RIS-Assisted UAV-Enabled MEC Systems," in *IEEE Transactions on Green Communications and Networking*, vol. 7, no. 4, pp. 1778-1792, Dec. 2023, doi: 10.1109/TGCN.2023.3287604.
- [32] J. Zhang, S. Chen, R. G. Maunder, R. Zhang and L. Hanzo, "Regularized Zero-Forcing Precoding-Aided Adaptive Coding and Modulation for Large-Scale Antenna Array-Based Air-to-Air Communications," in *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 9, pp. 2087-2103, Sept. 2018, doi: 10.1109/JSAC.2018.2864417.
- [33] E. Bobrov, B. Chinaev, V. Kuznetsov, et al., "Adaptive Regularized Zero-Forcing Beam Forming in Massive MIMO With Multi-Antenna Users," TechRxiv, 2021. doi: 10.36227/techrxiv.14892384.v1.
- [34] N. Xing, Y. Zhang, Y. Wang, and Y. Zhou, "Unmanned Aerial Vehicle Cooperative Data Dissemination Based on Graph Neural Networks," *Sensors* 24, no. 3: 887, 2024, doi: 10.3390/s24030887.
- [35] A. Kopic, E. Perenda and H. Gacanin, "A Collaborative Multi-Agent Deep Reinforcement Learning-Based Wireless Power Allocation with Centralized Training and Decentralized Execution," in *IEEE Transactions on Communications*, vol. 72, no. 11, pp. 7006-7016, Nov. 2024, doi: 10.1109/TCOMM.2024.3409530.
- [36] Y. Li, A. H. Aghvami and D. Dong, "Path Planning for Cellular-Connected UAV: A DRL Solution With Quantum-Inspired Experience Replay," in *IEEE Transactions on Wireless Communications*, vol. 21, no. 10, pp. 7897-7912, Oct. 2022, doi: 10.1109/TWC.2022.3162749.
- [37] M. Papini, M. Pirota, and M. Restelli, "Smoothing policies and safe policy gradients," *Machine Learning*, vol. 111, no. 3, pp. 4081-4137, 2022, doi: 10.1007/s10994-022-06232-6.
- [38] <https://www.kaggle.com/datasets/nigelwilliams/ngsim-vehicle-trajectory-data-us-101>, Accessed: July 2025.
- [39] S. Oubbati, A. Lakas, N. Lagraa and M. B. Yagoubi, "ETAR: Efficient Traffic Light Aware Routing Protocol for Vehicular Networks," 2015 International Wireless Communications and Mobile Computing Conference (IWCMC), Dubrovnik, Croatia, 2015, pp. 297-301, doi: 10.1109/IWCMC.2015.7289099.
- [40] X. Bai, B. Li, J. Li, Z. Wu, W. Zhang and S. S. Ge, "Efficient Routing for Multitruck Multidrone Package Delivery With Precedence Constraints," in *IEEE Transactions on Robotics*, vol. 41, pp. 6514-6528, 2025, doi: 10.1109/TRO.2025.3629808.
- [41] X. Bai, B. Li, I. Ullah, Z. Wu, S. Basheer, and A. K. Bashir, "Energy-efficient routing for IoT-enabled multi-truck multi-drone pickup and delivery systems," *Applied Energy*, vol. 400, p. 126546, Dec. 2025, doi: 10.1016/j.apenergy.2025.126546.